

1 Question (16 points)

Are the following languages regular? Prove your answer.

- For a given $p \in \mathbb{N}$, $A = \{a^n b^m \mid n, m \geq 0, n + m = p\}$
- $B = \{a^n b^m \mid 0 \leq n < m\}$

2 Question (16 points)

Are the following languages context-free? Prove your answer.

- $A = \{a^n b^m \mid n \neq m\}$
- $B = \{a^n b^m \mid m = n^3\}$

3 Question (18 points)

Design a TM that recognizes the language $L = \{a^n b^n c^n | n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

4 Question (16 points)

Describe a TM that recognizes the language $L = \{a^n b^{nm} | n, m \geq 1\}$. Do not design the TM. Describe the steps it needs to take.

5 Question (16 points)

Describe a multitape Turing machine with $k + 1$ tapes that recognizes the language $L = \{x_1^n x_2^n \dots x_k^n \mid n \geq 1\}$, $\Sigma = \{x_1, x_2, \dots, x_k\}$. You need to use all tapes in your solution. Do not design the TM. Describe the steps it needs to take.

6 Question (18 points)

Consider the language

$$L = \{ \langle D, k \rangle \mid D \text{ is DFA and it rejects all strings } w \text{ if } |w| < k \}, \text{ where } k \in \mathbb{N}$$

Is L decidable? Prove it.